



Jihyeon Son

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| PUBLICATIONS

- **12 hr Forecast of the SYM-H Index under Strong Southward Interplanetary Magnetic Field Conditions Using Deep Learning**
[Jihyeon Son](#), Yong-Jae Moon, Young-Sil Kwak, and Seungheon Shin
The Astrophysical Journal Supplement Series, 284(2), 46. (2026)
- **Six-hour Prediction of Interplanetary Magnetic Field Bz Profiles for Strong Southward Cases by Deep Learning**
[Jihyeon Son](#), Yong-Jae Moon, Young-Sil Kwak, Kyung Sun Park, and Hyun-Jin Jeong
The Astrophysical Journal, 984(1), 67. (2025)
- **Three-day Forecasting of Solar Wind Speed Using SDO/AIA Extreme-ultraviolet Images by a Deep-learning Model**
[Jihyeon Son](#), Suk-Kyung Sung, Yong-Jae Moon, Harim Lee, and Hyun-Jin Jeong
The Astrophysical Journal Supplement Series, 267(2), 45. (2023)
- **72-Hour Time Series Forecasting of Hourly Relativistic Electron Fluxes at Geostationary Orbit by Deep Learning**
[Jihyeon Son](#), Yong-Jae Moon, and Seungheon Shin
Space Weather, 20(10), e2022SW003153. (2022)
- **Generation of He I 1083 nm Images from SDO AIA Images by Deep Learning**
[Jihyeon Son](#), Junghun Cha, Yong-Jae Moon, Harim Lee, Eunsu Park, Gyungin Shin, and Hyun-Jin Jeong
The Astrophysical Journal, 920(2), 101. (2021)
- **Solar cycle Dependence of NOAA Space Weather Scale Frequencies**
Daeil Kim, Yong-Jae Moon, Hyun-Jin Jeong, [Jihyeon Son](#)
Journal of Korean Astronomical Society, 59(1), 55-61. (2025)

- 3D Magnetic Free Energy and Flaring Activity Using 83 Major Solar Flares**
Khojiakbar Karimov, Harim Lee, Hyun-Jin Jeong, Yong-Jae Moon, Jihye Kang, **Jihyeon Son**, Mingyu Jeon, Kanya Kusano
The Astrophysical Journal Letters, 965(1), L5. (2024)
- Application of Deep Learning to Solar and Space Weather Data**
Yong-Jae Moon, Harim Lee, **Jihyeon Son**, Suk-Kyung Sung, Kangwoo Yi, Hyun-Jin Jeong, Eunsu Park, Eun-Young Ji, Il-Hyun Cho, Bendict Lawrance, Daye Lim, Gyungin Shin, Sujin Lee, Sumiaya Rahman and Taeyoung Kim
Proceedings of the International Astronomical Union, 18, S372 (2023)

EDUCATION

2020.03 – 2024.02.

Combined Master’s and Doctoral Course in Solar Physics

School of Space Research, Kyung Hee University, Republic of Korea

2015.03 – 2020.02.

Bachelor of Science in Space science

Department of Astronomy & Space Science, Kyung Hee University, Republic of Korea

EMPLOYMENT

2025.05. –

Postdoctoral Researcher

Center for Heliophysics Research, Korea Astronomy & Space Science Institute, Republic of Korea

2024.03. – 2025.04.

Postdoctoral Researcher

Astronomy & Space Science, College of Applied Science, Kyung Hee University, Republic of Korea

PATENTS

2025.05.

- Method, apparatus and system for predicting southward Component of interplanetary magnetic field using BILSTM
Yong-Jae Moon and **Jihyeon Son**
10-2025-0059892, Republic of Korea Patent Application
- Apparatus for predicting Solar Wind Speed using Deep Learning Model and Method thereof
Yong-Jae Moon and **Jihyeon Son**

2023.07.

AWARDS	10-2023-0087337, Republic of Korea Patent Application	2026.09.
	Outstanding Postdoctoral Researcher Award in Korea Astronomy and Space Science Institute	
		2020.12.
PARTICIPATED PROJECTS	Grand Prize, Korean Space Weather Center: Artificial Intelligence (AI) competition for space weather forecasting	
	Development of prediction models for solar wind parameters and geomagnetic activity using deep learning Role: Principal Investigator	2024.05. – present
	Study on the solar wind forecast and 3D ionospheric modelling improvement using deep learning Role: Development of solar wind forecasting model	2023.01. – 2023.12.
	Study on the forecast of solar winds and IGS 3D ionospheric modelling technique using deep learning Role: Development of solar wind speed forecasting model	2023.01. – 2023.12.
	Development of analysis and forecast models for space weather operations Role: Development of space weather forecasting model	2021.12. – 2024.12.
	Study on the forecast of the occurrence, strength, and temporal evolution of solar flares using deep learning Role: Development of flare forecasting model	2020.05 – 2022.02
SCHOOL PROGRAM	Python in Heliophysics Summer School <i>Madrid, Spain</i>	2022.05.
TECHNICAL REPORTS	- Python - Analysis of solar image data and solar wind data - Deep learning: Tensorflow keras & Pytorch - Microsoft (Word, Excel, Powerpoint)	